

## 2.1 Results

| Label | N min | T min(ns) | N max(ns) | T max(ns) |
|-------|-------|-----------|-----------|-----------|
| QS    | 10    | 3200      | 100       | 162300    |
| QR    | 10    | 2400      | 100       | 105500    |
| QC    | 10    | 3200      | 100       | 186700    |
| MS    | 10    | 2900      | 100       | 67000     |
| MR    | 10    | 4600      | 100       | 57900     |
| MC    | 10    | 3600      | 100       | 43000     |
| IS    | 10    | 600       | 100       | 2900      |
| IR    | 10    | 500       | 100       | 3100      |
| IC    | 10    | 700       | 100       | 3200      |
| SS    | 10    | 1300      | 100       | 102800    |
| SR    | 10    | 1400      | 100       | 90900     |
| SC    | 10    | 2000      | 100       | 98300     |

## 2.2 Analysis

| Label | Tmax/Tmin | N ratio | N ln n ratio | N <sup>2</sup> ratio | Behavior  |
|-------|-----------|---------|--------------|----------------------|-----------|
| QS    | 51        | 10      | 20           | 100                  | N log n   |
| QR    | 44        | 10      | 20           | 100                  | N log n   |
| QC    | 58        | 10      | 20           | 100                  | N log n   |
| MS    | 23        | 10      | 20           | 100                  | N log n   |
| MR    | 13        | 10      | 20           | 100                  | Linear    |
| MC    | 12        | 10      | 20           | 100                  | Linear    |
| IS    | 5         | 10      | 20           | 100                  | Linear    |
| IR    | 6         | 10      | 20           | 100                  | Linear    |
| IC    | 5         | 10      | 20           | 100                  | Linear    |
| SS    | 79        | 10      | 20           | 100                  | Quadratic |
| SR    | 65        | 10      | 20           | 100                  | Quadratic |
| SC    | 49        | 10      | 20           | 100                  | N log n   |

## Summary:

Overall, my results mostly matched what I expected from theory. Insertion sort and selection sort clearly showed quadratic behavior since their times increased a lot as  $N$  got bigger. Merge sort and quicksort both followed the expected  $O(n \log n)$  trend and were much faster overall. One thing that actually surprised me was that merge sort sometimes ran faster than quicksort, even though quicksort is usually considered the faster algorithm. I think this happened because merge sort is more consistent, while quicksort's performance can vary depending on how the pivots are chosen. I also noticed some random differences in times between runs, probably because I had other apps open like Spotify, VS Code, and Chrome. Even with those small variations, the overall results made sense and showed the expected patterns for each sorting algorithm.